$$\underset{(A)}{Na_2S_2O_3} + 2HCl \longrightarrow 2NaCl + \underset{(B)}{SO_2}\uparrow + \underset{\text{yellow ppt}}{S}\downarrow + H_2O$$

$$K_2Cr_2O_7 + H_2SO_4 + 3SO_2 \longrightarrow K_2SO_4 + \underset{(C)}{Cr_2(SO_4)_3} + H_2O$$

15. a) $BrO_3^-$ , $MnO_2$

$$2MnO_4^- + Br^- + H_2O \longrightarrow 2OH^- + 2MnO_2 + BrO_3^-$$

16. c) 3

$$K_2Cr_2O_7 + 6KI + 7H_2SO_4 \longrightarrow 4K_2SO_4 + Cr_2(SO_4)_3 + 7H_2O + 3I_2$$

17. c) 0.6

$$\overset{7+}{Mn}O_4^- + \overset{+2\ +3}{Fe}C_2O_4 \longrightarrow Mn^{2+} + Fe^{3+} + 2\overset{+4}{C}O_2$$

$5e^-$ acception  $3e^-$ release

5 moles of $FeC_2O_4 \equiv$ 3 moles of $KMnO_4$

1 mole of $FeC_2O_4 \equiv \left(\frac{3}{5}\right)$ moles of $KMnO_4$

1 mole of $FeC_2O_4 \equiv$ 0.6 moles of $KMnO_4$

18. a) $MnO_2$, $Cl_2$ , $NCl_3$

$$\underset{(A)}{MnO_2} + 4HCl \longrightarrow MnCl_2 + \underset{(B)}{Cl_2}\uparrow + H_2O$$

$$NH_3 + 3Cl_2 \longrightarrow \underset{(C)}{NCl_3} + 3HCl$$

19. c) All the lanthanons are much more reactive than aluminium.

As we move from La to Lu , their metallic behaviour because almost similar to that of aluminium.

20. b) $Yb^{2+}$

$Yb^{2+}$ - $4f^{14}$ –no unpaired electrons - diamagnetic

21. d) 3

22. a) Both assertion and reason are true and reason is the correct explanation of assertion.

23. c) +4

24. a) Np, Pu ,Am

25. a) $La(OH)_2$ is less basic than $Lu(OH)_3$



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